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BIOMEDICAL LIMITED

INSTALLATION QUALIFICATION (IQ)

PROTOCOL
FOR
AIR SHOWER

EQUIPMENT SR. NO. –

EQUIPMENT ID. –

LOCATION – INFUSION FACTORY FILLING ROOM

PROTOCOL No. – BML/PDF/FR/PT/26/001

PROVIDED BY:

BIOMEDICAL LIMTED

KWARA STATE, NIGERIA

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1.0. PROTOCOL FORMAT APPROVAL:

Signatures in this page indicate that this document and its format have been prepared, checked and approved by the personnel concerned. It shall be signed and dated as shown below.

	WRITTEN BY:	**REVIEWED BY:**
Signature		
Position		
Date		

	APPROVED BY:	AUTHORISED BY:
Signature		
Position		
Date		

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2.0 OBJECTIVE

The purpose of the study is to ensure proper installation of the four **AIR SHOWER** at **THE FILLING ROOM, INFUSION FACTORY, BIOMEDICAL LIMITED.**

This study verifies:

- Qualification of the equipment, re installation and to ensure that all background information is performed to assure conformance to **BIOMEDICAL LIMITED** and manufacturer's specifications.
- The equipment meets the current good manufacturing practices requirements and all other regulatory obligations
- No unauthorized or unrecorded modifications have taken place.
- Operation procedures/preventive maintenance procedures have been identified and listed.
- All critical instrumentation has been identified
- All utilities are properly connected
- All safety features are accounted for
- All relevant data gave have been checked

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3.0 SCOPE

- This protocol is applicable in carrying out the installation qualification of the Air Shower at the Infusion manufacturing site, Biomedical Limited, Kwara State.
- To be performed after the completion of the design qualification
- To be performed at the time of relocation.

4.0 RESPONSIBILITIES AND QUALIFICATION TEAM

The following departments shall be involved in:

RESPONSIBILITY	DEPARTMENT
Provision of needed utility and machine installation	Engineering Team
Review of the Installation Qualification Document	Head Of Engineering
Final Documentation and system acceptability	QA –Manager
Authorization	Superintendent Pharmacist/factory Manager

The following personnel will be involved in the execution of the Qualification

S/No.	DEPARTMENT	NAME	SIGNATURE
1			
3			
4			
5			

5.0 EQUIPMENT IDENTIFICATION AND DESCRIPTION

IDENTIFICATION

Equipment Name	AIR SHOWER
Manufacturer's Name	SHANGAI IVEN PHAMATECH ENGINEERING CO. LTD (CHINA)
Supplier's name	SHANGAI IVEN PHAMATECH ENGINEERING CO. LTD (CHINA)
Location of Installation	FILLING ROOM (INFUSION FACTORY)

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DESCRIPTION:

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6.0 PREREQUISITE

The equipment shall confirm the specifications and requirements as specified in the Design Qualification

S/N	ACCEPTANCE CRITERIA
1	User Requirement specification
2	Design Qualification

7.0 ABBREVIATIONS AND DEFINITIONS

Abbreviation	Definition
SOP	Standard Operating Procedure
MOC	Material of Construction
HMI	Human Machine Interface
ENGR	Engineering
IQ	Installation Qualification
QA	Quality Assurance
VAC	Voltage Alternating Current
S. S	Stainless Steel
LTD	Limited
V	Voltage
GMP	Good manufacturing practices
AC	Alternating Current
Amps	Ampere
W	Watt
SMPS	Switched mode power supply
VDC	Voltage Direct Current
DC	Direct Current
PVC	Polyvinyl Chloride
MCB	Miniature Circuit Breaker

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8.0 REFERENCES:

The list of reference documents, which are referred for preparation and review of this document, is as follows: -

- S.O.P for Installation, Operation & Performance Qualification of equipment & Instrument: BML-QDG-QA-111
- Machine Manual supplied by SHANGAI IVEN PHAMATECH ENGINEERING CO. LTD (CHINA)
- Design Qualification supplied by SHANGAI IVEN PHAMATECH ENGINEERING CO. LTD (CHINA)

9.0 EQUIPMENT INSTALLATION CHECKLIST:

Pre-Installation Consideration

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	There is no observable physical damage to the equipment surface and all the contact parts are free from scratches & damages.			
2	Sufficient room for servicing, handling & maintenance is provided (around the machine)			
3	Check the floor space. It should not be inclined.			

Utilities

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Power Requirement (AC380V/50Hz/2.5KW)			
2	Adequate illumination at the Proposed Machine area			
3	Proper Earthing System available			

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Equipment compliance

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Air Shower name plate present			
2	Air Shower Dimension (Outer diameter: W1290*D1500*H2180mm, Inner diameter: W790*D1360*H1930mm)			
3	Air Shower Model No. present (FLS-1290-1500-2180)			
4	MOC: SUS304			
5	Air Shower Serial Number Present (FLS2402430-05/ FLS2402430-06/ FLS2402430-07/ FLS2402430-08)			

Safety compliance

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	All external surfaces are free of sharp edges.			
2	Light indicator is available			
3	Emergency Stop Buttons available			

Electrical diagram check

S/N	Description	Qualified / Not Qualified	Checked By	Verified By
1	Verify the electrical panel wiring diagram			

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2	Check whether the position of each component in the electrical is correct			
3	Check whether the wiring of the electrical panel is consistent with electrical diagrams			
4	Check if the electrical diagram is attached to the operation manual			

List of Main Components with their material of construction

S/N	Component	Origin/ Configuration	Qualified / Not Qualified	Checked By	Verified By
1	Main Frame	IVEN China, S.S 304			
2	Centrifugal blower	Model: EDF-240, Motor power: 1.1kW, three phase, Voltage: 380/50Hz, Blade material: SS, Traffic: 2550m^3/hr., Rotational Speed: 2840r/min			
3	Air nozzles	18 pieces (Adjustable), MOC: SS304			
4	Pre-filter	Model: G4			
5	Lighting	18w Flush LED Bulb (square)			
6	HEPA	IVEN China, Class: H14 Filter Efficiency: 99.99%@MPPS, Particle Size: 0.3µm Dimension: 610X610X100mm			
7	Air Interlock	Electromagnetic lock			
8	Door	Manual swing door, transparent tempered glass type, with door closer			
9	Emergency button	220vac			
10	Infrared Sensor	220VAC			
11	Air shower control panel	China, Model: FLS2A18-L01 V1.0			
12	Circuit Breaker	Manufacturer: Chint, Model: NXB-63 D25. 400V, 50HZ 6000A. 4- Poles			
13	Contactor	Manufacturer: Chint, Model: CJX2-1210			

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10.0 INSTALLATION QUALIFICATION REPORT:

Attachment of all reports written as:

#Attachment 1

#Attachement 2 etc.

Date of Installation:

Date of Completion:

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11.0 DEVIATION REPORT

Prepare a Deviation report including the justification of acceptance and impact on the function of the system/equipment

DEVAITIONS:
JUSTIFICATION FOR ACCEPTANCE:
IMPACT ON OPERATION:

Written By: **Date:**

Reviewed By: **Date:**

Approved By: **Date:**

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12.0 CONCLUSION

13.0 List of Attachment

Attach all completed, signed data record forms

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14.0 REPORT APPROVAL

Written By: **Date:**

Reviewed By: **Date:**

Approved By: **Date:**

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